

1.5

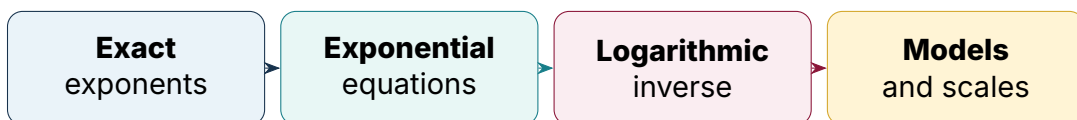
Exponent Laws and Logarithms

Full Notes - Exact Exponents - Exponential Equations - Logarithms - Models - TI-84

Central idea. Exponents describe repeated multiplicative change. Logarithms reverse that change by answering one question: **what exponent produces this value?**

$$b^y = x \iff \log_b x = y$$

$$x = \frac{\log c}{\log b} \text{ when } b^x = c$$



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Version: Student Full Notes

Learning route and formula map

exact structure → inverse operation → verify → interpret

Learning intentions

By the end of Lesson 1.5, you should be able to:

1. simplify integer, zero, negative, and rational exponents exactly and state restrictions;
2. solve exponential equations by common bases, substitution, logarithms, or graph intersection;
3. translate fluently between $b^y = x$ and $\log_b x = y$;
4. use logarithm laws and change of base accurately;
5. solve growth, decay, and threshold problems and distinguish continuous crossings from discrete recorded times;
6. interpret logarithmic scales, including sound intensity and pH;
7. use TI-84 evidence transparently and verify calculator output independently;
8. discuss domain, units, assumptions, and model limitations.

Core reference sheet

Product	$a^m a^n = a^{m+n}$
Quotient	$\frac{a^m}{a^n} = a^{m-n}, a \neq 0$
Power of a power	$(a^m)^n = a^{mn}$
Power of a product	$(ab)^n = a^n b^n$
Zero exponent	$a^0 = 1, a \neq 0$
Negative exponent	$a^{-n} = \frac{1}{a^n}, a \neq 0$
Rational exponent	$a^{m/n} = \sqrt[n]{a^m}$ where the real expression is defined
Log definition	$\log_b x = y \iff b^y = x, \text{ with } b > 0, b \neq 1, x > 0$
Log product law	$\log_b(MN) = \log_b M + \log_b N$
Log quotient law	$\log_b(M/N) = \log_b M - \log_b N$
Log power law	$\log_b(M^p) = p \log_b M$
Change of base	$\log_b x = \frac{\log x}{\log b} = \frac{\ln x}{\ln b}$

Calculator principle

GDC NOT NEEDED Exact algebra should remain exact whenever practical. **GDC EXPECTED** Non-integer logarithmic solutions, logarithmic-scale calculations, and numerical intersections are natural calculator tasks. **GDC SKILL** Always record the model entered, calculator output, an independent check, and the contextual conclusion.

1.5A - Exponent Laws and Exact Powers

structure before buttons

What an exponent records

For a positive integer n ,

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ equal factors}}.$$

Exponent laws compress repeated multiplication. They are rules about **products and quotients**; they do not create a rule such as $(a + b)^n = a^n + b^n$.

Product law - the corrected visual idea

The expression $a^3 a^4$ means **multiplication** of two powers:

$$\underbrace{a \cdot a \cdot a}_{3 \text{ factors}} \times \underbrace{a \cdot a \cdot a \cdot a}_{4 \text{ factors}} = \underbrace{a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a}_{7 \text{ factors}} = a^7.$$

The *factor counts* add: $3 + 4 = 7$. The powers themselves are not being added.

Worked Example 1

GDC NOT NEEDED

Simplify $6x^4 y^{-2} \cdot 3x^{-1} y^5$, writing the final answer with positive exponents.

WORKING / ANNOTATIONS

Your Turn 1

GDC NOT NEEDED

Simplify $4a^{-3}b^2 \cdot 5a^7b^{-5}$ with positive exponents.

WORKING / ANNOTATIONS

Quotient and power laws

$$\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0), \quad (a^m)^n = a^{mn}, \quad (ab)^n = a^n b^n.$$

For example,

$$\frac{12x^7y^{-2}}{3x^{-1}y^4} = 4x^{7-(-1)}y^{-2-4} = \frac{4x^8}{y^6}.$$

Common trap

The exponent law applies to factors: $(ab)^n = a^n b^n$. It does **not** distribute across addition. For example, $(1+1)^2 = 4$ but $1^2 + 1^2 = 2$.

Zero exponent

For $a \neq 0$,

$$\frac{a^m}{a^m} = 1 \quad \text{and} \quad \frac{a^m}{a^m} = a^{m-m} = a^0.$$

Therefore $a^0 = 1$ for $a \neq 0$. The restriction matters because the proof divides by a^m .

Negative exponents

For $a \neq 0$,

$$a^{-n} = \frac{1}{a^n}$$

A negative exponent does not make the whole value negative; it indicates a reciprocal.

$$5^{-2} = \frac{1}{25}, \quad \left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}.$$

Worked Example 2

GDC NOT NEEDED

Simplify $\frac{(2x^{-2}y^3)^2}{4x^{-1}y}$ with positive exponents.

WORKING / ANNOTATIONS

Your Turn 2

GDC NOT NEEDED

Evaluate exactly: (a) $64^{-2/3}$, (b) $\left(\frac{64}{125}\right)^{2/3}$, (c) $16^{-3/4}$.

WORKING / ANNOTATIONS

Rational exponents

For a real expression where the required root exists,

$$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$$

Examples:

$$27^{2/3} = (\sqrt[3]{27})^2 = 9, \quad 16^{3/4} = (\sqrt[4]{16})^3 = 8.$$

Under the usual real-number interpretation, an even root of a negative number is not real, while an odd root may be real. A negative rational exponent also requires the resulting denominator to be non-zero.

Domain and restriction audit

Before accepting a simplified expression, ask:

- Did the original expression contain a denominator that cannot be zero?
- Did a negative exponent introduce a reciprocal?
- Does an even root require a non-negative radicand?
- Are all restrictions from the *original* expression still recorded?

Checkpoint - Exact exponent structure

1. Simplify $\frac{(a^4)^3}{a^5}$ for $a \neq 0$.
2. Evaluate $32^{-3/5}$ exactly.
3. Explain why $(-8)^{1/2}$ is not real but $(-8)^{1/3}$ is real.
4. Decide whether $(x + y)^4 = x^4 + y^4$ is an identity and justify your decision.

WORKING / ANNOTATIONS

1.5B - Exponential Equations and Common Bases

exact route first

Decision tree for exponential equations

1. **Can both sides use one base?** Rewrite and equate exponents.
2. **Can you factor a common exponential term?** Factor before using logarithms.
3. **Is there a repeated expression such as 3^x and 9^x ?** Substitute $u = 3^x > 0$.
4. **Otherwise, isolate the exponential expression and take logarithms.**
5. If an equation mixes unlike functions, a graph intersection may be the efficient numerical route.

Worked Example 3

GDC NOT NEEDED

Solve $9^{x-1} = 3^{x+3}$.

WORKING / ANNOTATIONS

Your Turn 3

GDC NOT NEEDED

Solve: (a) $4^{3x-1} = 8^{x+2}$; (b) $25^{x+1} = 5^{3x-2}$.

WORKING / ANNOTATIONS

Identity and no-solution cases

Common-base rewriting can reveal special cases immediately.

$$2^{3x-1} = 8^{x+2} = 2^{3x+6}$$

would require $3x - 1 = 3x + 6$, impossible, so there is **no solution**. By contrast,

$$4^{x+1} = 2^{2x+2}$$

is true for **all real** x because the two sides are identical after rewriting.

Worked Example 4

GDC NOT NEEDED

Solve $2^x + 2^{x+1} = 96$.

WORKING / ANNOTATIONS

Worked Example 5

GDC NOT NEEDED

Solve $9^x - 10 \cdot 3^x + 9 = 0$.

WORKING / ANNOTATIONS

Your Turn 4

GDC NOT NEEDED

Solve $4^x + 2^{x+1} - 8 = 0$ using a substitution.

WORKING / ANNOTATIONS

When logarithms are necessary

If $b^x = c$ with $b > 0$, $b \neq 1$, $c > 0$, then

$$x \ln b = \ln c \implies x = \frac{\ln c}{\ln b}.$$

Use full calculator precision until the final reported answer.

Worked Example 6

GDC EXPECTED

Solve $3^{2x-1} = 17$, giving x to three decimal places.

WORKING / ANNOTATIONS

Your Turn 5

GDC EXPECTED

Solve $5(2^x) = 70$, giving x to three decimal places, and verify.

WORKING / ANNOTATIONS

Graph intersection as a numerical method

For an equation such as $2^x = 5x$, neither side can be reduced to the same simple base and there may be more than one intersection. Graphing both sides is useful, but the **window must be wide enough to reveal all crossings**.

Worked Example 7

GDC SKILL

Solve $2^x = 5x$ numerically.

WORKING / ANNOTATIONS

Checkpoint - Exponential equations

1. Solve $8^{x-1} = 4^{x+2}$.
2. Solve $2^{3x-1} = 8^{x+2}$ and classify the solution set.
3. Solve $4^{x+1} = 2^{2x+2}$ and classify the solution set.
4. Explain why equating exponents in $3^x = 5^{x-1}$ is invalid.

WORKING / ANNOTATIONS

1.5C - Logarithms and Model Inversion

inverse multiplicative change

Definition of a logarithm

For $b > 0$, $b \neq 1$, and $x > 0$,

$$\log_b x = y \iff b^y = x$$

The **base** is b , the positive **argument** is x , and the logarithm value is the exponent y .

$$2^5 = 32 \iff \log_2 32 = 5, \quad 10^{-3} = 0.001 \iff \log_{10}(0.001) = -3.$$

Real logarithm conditions

A real logarithm requires:

$$b > 0, \quad b \neq 1, \quad x > 0.$$

The argument condition $x > 0$ is not optional. A positive real base raised to any real power remains positive, so it cannot produce zero or a negative argument.

Your Turn 6

GDC NOT NEEDED

Rewrite each statement in the other form.

(a) $3^4 = 81$ (b) $\log_5(1/125) = -3$ (c) $e^2 = x$.

WORKING / ANNOTATIONS

Common and natural logarithms

On a TI-84:

$$\log x = \log_{10} x, \quad \ln x = \log_e x.$$

Both are valid for change of base, provided the same logarithm base is used in numerator and denominator.

Logarithm laws

For positive M, N and valid base b ,

$$\log_b(MN) = \log_b M + \log_b N,$$

$$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N, \quad \log_b(M^p) = p \log_b M.$$

These are consequences of exponent laws. There is **no** corresponding law for $\log(M + N)$.

Worked Example 8**GDC NOT NEEDED**

Given $\log A = 2.3$ and $\log B = 0.8$, find (a) $\log(AB)$, (b) $\log(A/B)$, (c) $\log(A^3)$.

WORKING / ANNOTATIONS**Common trap**

$\log(20 + 30) \neq \log 20 + \log 30$. Numerically, the right side minus the left side is about 1.079181, not 0. Product, quotient, and power laws do not become a sum law.

Change of base

For any convenient logarithm base,

$$\log_b x = \frac{\log x}{\log b} = \frac{\ln x}{\ln b}$$

Example:

$$\log_2(19) = \frac{\log 19}{\log 2} \approx 4.247927513.$$

Verification: $2^{4.247927513} \approx 19$.

Your Turn 7**GDC EXPECTED**

Evaluate to three decimal places: (a) $\log_7(100)$, (b) $\log_3(20)$, (c) $\log_{1/2}(8)$.

WORKING / ANNOTATIONS

Logarithmic equations - domain first

When solving a logarithmic equation:

1. state all argument restrictions first;
2. combine logs only when the laws apply;
3. convert to exponential form or solve algebraically;
4. test every algebraic candidate in the **original** logarithms.

Worked Example 9

GDC NOT NEEDED

Solve $\log_{10}(x - 1) + \log_{10}(x - 3) = 1$.

WORKING / ANNOTATIONS

Your Turn 8

GDC NOT NEEDED

Solve: (a) $\log_2(x - 1) = 3$; (b) $\log_{10} x + \log_{10}(x - 9) = 1$; (c) $2 \ln x = \ln 49$.

WORKING / ANNOTATIONS

Domains of logarithmic expressions

The entire logarithm argument must be positive.

$$\log(2x - 10) : 2x - 10 > 0 \Rightarrow x > 5.$$

For a quotient or product, solve a sign inequality rather than requiring each factor separately positive. For example,

$$\log\left(\frac{x - 1}{x + 2}\right)$$

requires $(x - 1)/(x + 2) > 0$, giving

$$(-\infty, -2) \cup (1, \infty).$$

Exponential models and threshold inversion

GDC EXPECTED

Discrete exponential model

A constant percentage change per recorded period has form

$$Q(t) = Q_0 b^t$$

where Q_0 is the initial value and $b = 1 + r$ is the factor per time unit. If $b > 1$, the model grows; if $0 < b < 1$, it decays.

Worked Example 10

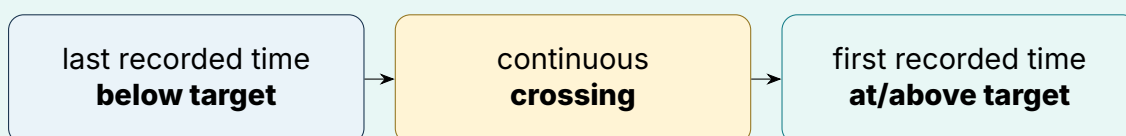
GDC EXPECTED

A culture follows $N(t) = 650(1.14)^t$. Determine when it reaches 3000 cells. If observations occur only at whole hours, state the first recorded hour above 3000.

WORKING / ANNOTATIONS

Three-stage threshold evidence

For a discrete increasing model, show all three pieces:



A decimal time is not automatically an admissible discrete stage.

Your Turn 9

GDC EXPECTED

For $P(t) = 12000(0.97)^t$, find the first whole year for which $P < 9000$. Show adjacent-value evidence.

WORKING / ANNOTATIONS

Continuous exponential model

A continuous model is often written

$$Q(t) = Q_0 e^{kt}$$

where k has reciprocal-time units. If $k > 0$ the model grows; if $k < 0$ it decays. To solve for time,

$$t = \frac{\ln(Q/Q_0)}{k}.$$

Worked Example 11

GDC EXPECTED

A quantity doubles every 7.5 years under $Q = Q_0 e^{kt}$. Find k .

WORKING / ANNOTATIONS

Doubling time and half-life

For discrete factor b :

$$T_{\text{double}} = \frac{\ln 2}{\ln b} \quad (b > 1), \quad T_{1/2} = \frac{\ln(1/2)}{\ln b} \quad (0 < b < 1).$$

For continuous $Q = Q_0 e^{kt}$:

$$T_{\text{double}} = \frac{\ln 2}{k} \quad (k > 0), \quad T_{1/2} = \frac{\ln(1/2)}{k} \quad (k < 0).$$

Your Turn 10

GDC EXPECTED

Find (a) the doubling time at 4.8% annual growth, and (b) the half-life when a value decreases by 12% per year.

WORKING / ANNOTATIONS

Checkpoint - Logarithms and inversion

1. State the real domain of $\ln(7 - 3x)$.
2. Solve $240(1.09)^t = 900$ to two decimal places.
3. Explain why a decreasing model can still produce a positive time from a quotient of logarithms.
4. A continuous crossing occurs at $t = 12.23$ days and measurements are whole days. State the first recorded day below the threshold, with the needed verification statement.

WORKING / ANNOTATIONS

1.5D - Logarithmic Scales, Modelling, and Mastery

ratios hidden inside differences

What a logarithmic scale does

A logarithmic scale compresses a large multiplicative range. Equal distances on the scale represent equal **ratios**, not equal differences. On a base-10 log axis, 10, 100, and 1000 are equally spaced because each step multiplies by 10.

Sound intensity level

A common model is

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right) \text{ dB}$$

A difference ΔL in decibels corresponds to intensity ratio

$$\frac{I_2}{I_1} = 10^{\Delta L/10}$$

Thus +10 dB means $\times 10$ intensity, +20 dB means $\times 100$, and +30 dB means $\times 1000$.

Worked Example 12

GDC EXPECTED

Two sound levels differ by 18 dB. Find the intensity ratio.

WORKING / ANNOTATIONS

Your Turn 11

GDC EXPECTED

Sound intensity is 5.0×10^6 times the reference intensity. Find $L = 10 \log_{10}(I/I_0)$.

WORKING / ANNOTATIONS

Acidity and pH

$$\text{pH} = -\log_{10}[H^+]$$

where $[H^+] > 0$. The negative sign reverses direction: lower pH means larger hydrogen-ion concentration. A decrease of 1 pH unit multiplies $[H^+]$ by 10.

Worked Example 13

GDC EXPECTED

A solution has pH 5.6. Find $[H^+]$ in mol dm^{-3} .

WORKING / ANNOTATIONS

Your Turn 12

GDC EXPECTED

Solution A has pH 6 and solution B has pH 4. How many times greater is $[H^+]$ in B?

WORKING / ANNOTATIONS

Recovering an exponential model from data

If $Q(t) = Ab^t$ and two points are known, dividing the equations eliminates A . For $Q(2) = 400$ and $Q(7) = 900$,

$$\frac{900}{400} = b^5 = 2.25, \quad b = 2.25^{1/5} \approx 1.176079.$$

Then

$$A = \frac{400}{b^2} \approx 289.192.$$

The factor $b \approx 1.1761$ means about 17.61% growth per time unit.

Worked Example 14

GDC EXPECTED

Using $Q(2) = 400$ and $Q(7) = 900$, determine when $Q = 1500$.

WORKING / ANNOTATIONS

Log-linear form and residuals

If $y = Ab^t$, then

$$\log_{10} y = \log_{10} A + t \log_{10} b.$$

So a plot of $\log_{10} y$ against t should be linear for an exact exponential model. A systematic curve in the residuals suggests the growth factor is not constant across the whole interval; a different or piecewise model may be more appropriate.

Model validity checklist

A mathematically correct exponential calculation is not automatically a good prediction. Check:

- the time units and domain;
- whether the factor/rate can reasonably remain constant;
- whether capacity, resources, treatment changes, or external conditions create saturation;
- whether discrete measurements require integer-stage reporting;
- whether residuals show random variation or systematic model failure.

TI-84 Classroom Training - Five Required Workflows

MODEL → ENTER → READ → INTERPRET → VERIFY

TI-84 C1 - LOG / LN / change of base

GDC SKILL

Problem: Evaluate $\log_2(19)$ and explain why LOG and LN both work.

Manual model:

$$\log_2(19) = \frac{\log 19}{\log 2} = \frac{\ln 19}{\ln 2}.$$

Keys 1: LOG 19) ÷ LOG 2) ENTER

Keys 2: LN 19) ÷ LN 2) ENTER

Output: 4.247927513...

VERIFY: $2^{4.247927513} \approx 19$.

IB conclusion: $\log_2(19) \approx 4.24793$; LOG and LN agree because change of base uses the same base in numerator and denominator.

TI-84 C2 - Solve an exponential equation

GDC SKILL

Problem: Solve $3^x = 17$.

Manual model: $x = \ln 17 / \ln 3$. **Keys:** LN 17) ÷ LN 3) ENTER

Output: $x \approx 2.578901923$.

VERIFY: $3^{2.578901923} \approx 17$.

IB conclusion: $x \approx 2.579$, verified by substitution.

TI-84 C3 - Find both graph intersections**GDC SKILL**

Problem: Solve $2^x = 5x$ and ensure the graph window does not hide a solution.

Y=: Enter $Y_1 = 2^X$ and $Y_2 = 5X$.

WINDOW: For example, use an x -window that includes 0 through at least 6 and a suitable y -range.

GRAPH: Count visible crossings before solving.

Intersect: 2nd TRACE 5:intersect; repeat with a guess near each crossing.

Outputs: $x \approx 0.2354557101$ and $x \approx 4.488001136$.

VERIFY: Substitute both values into 2^x and $5x$.

IB conclusion: There are two real solutions, $x \approx 0.23546$ and $x \approx 4.48800$.

TI-84 C4 - Logarithm domain rejection**GDC SKILL**

Problem: Solve $\log(x - 1) + \log(x - 3) = 1$.

Domain first: $x > 3$.

Algebra: $(x - 1)(x - 3) = 10$, so $x = 2 \pm \sqrt{11}$.

Calculator support: use the home screen to evaluate $2 + \sqrt{11} \approx 5.31662479$ and, if desired, test both algebraic candidates.

VERIFY: $2 - \sqrt{11} < 3$ is invalid; only $2 + \sqrt{11}$ gives positive arguments.

IB conclusion: $x = 2 + \sqrt{11} \approx 5.31662$ is the only real solution.

TI-84 C5 - Decay threshold with integer-stage check**GDC SKILL**

Problem: Find the first integer n for which $84(0.78)^n < 15$.

Continuous crossing: $\text{LOG}(15/84) \div \text{LOG}(0.78)$ ENTER gives $n \approx 6.93374$.

Y=: Enter $Y_1 = 84(0.78)^X$.

TABLE: Inspect $n = 6$ and $n = 7$.

Outputs: $P_6 \approx 18.9168$, $P_7 \approx 14.7551$.

VERIFY: $P_6 \geq 15$ and $P_7 < 15$.

IB conclusion: The first integer stage below 15 is $n = 7$.

TI-84 habits that prevent lost marks

- Do not round individual logarithms before dividing them.
- For graph intersections, count crossings and choose a window before using Intersect.
- A decimal threshold time locates the boundary; adjacent admissible stages prove the final discrete answer.
- A calculator cannot repair a missing logarithm domain restriction.
- Write enough of the model and method that another reader can reproduce your result.

IB-style Mixed Practice

communicate complete mathematical evidence

IB Task A - Exact exponent reasoning**GDC NOT NEEDED**

Let $a, b \neq 0$ where all stated rational powers are real.

- (a) Simplify $\frac{(a^3b^{-2})^2a^{-1}}{a^2b^{-5}}$ with positive exponents. [3]
- (b) Evaluate $32^{-3/5}$ exactly. [2]
- (c) Solve $9^{x-1} = 27^{2-x}$. [3]
- (d) Disprove $(p + q)^3 = p^3 + q^3$ with a counterexample. [2]

WORKING / ANNOTATIONS

IB Task B - Culture threshold**GDC EXPECTED**

A culture is modelled by $N(t) = 650(1.14)^t$, with t in hours.

- (a) Calculate $N(6)$ to the nearest whole cell. [2]
- (b) Determine when the model reaches 3000 cells. [3]
- (c) Observations occur at whole hours. Find the first observation time above 3000. [2]
- (d) Interpret the factor 1.14. [2]
- (e) State one limitation of extending the model indefinitely. [2]

WORKING / ANNOTATIONS

IB Task C - Sound intensity**GDC EXPECTED**

Sound level is $L = 10 \log_{10}(I/I_0)$ dB.

- (a) Find L when $I = 100I_0$. [2]
- (b) Find I/I_0 when $L = 35$ dB. [3]
- (c) Two sounds differ by 18 dB. Find their intensity ratio. [2]
- (d) Explain why equal additions to L correspond to multiplicative intensity changes. [3]
- (e) State the domain restrictions on I and I_0 . [2]

WORKING / ANNOTATIONS

IB Task D - Reconstruct an exponential model**GDC EXPECTED**

A quantity is modelled by $Q(t) = Ab^t$, with $Q(2) = 400$ and $Q(7) = 900$.

- (a) Show that $b^5 = 2.25$. [2]
- (b) Find b to four decimal places. [2]
- (c) Find A to three significant figures. [2]
- (d) Determine when $Q = 1500$. [3]
- (e) Interpret b . [1]
- (f) State one check that would help assess reliability beyond the two measured points. [2]

WORKING / ANNOTATIONS**Final Mastery Check**

exact - numerical - contextual

Checkpoint - Mixed mastery

1. Simplify $\frac{12x^7y^{-2}}{3x^{-1}y^4}$ with positive exponents.
2. Solve $2^x + 2^{x+1} = 96$.
3. Evaluate $\log_4(1/64)$.
4. Solve $\log_3 x + \log_3(x - 2) = 2$, respecting the domain.
5. Two sound levels differ by 15 dB. Find the intensity ratio.
6. A decreasing model crosses a threshold at 12.23 days; observations are whole days. State the first recorded day below the threshold and what must be checked.

WORKING / ANNOTATIONS

What a strong IB response looks like

1. Write the exact structure or model first.
2. State restrictions before solving logarithmic expressions.
3. Use exact algebra where it is efficient.
4. When the GDC is appropriate, record the entered expression or graph functions.
5. Keep guard digits until the final answer.
6. Verify by substitution, a second representation, or adjacent-stage evidence.
7. End with units and contextual meaning, not only a number.

Exit reflection

Complete these statements in your own words:

- The exponent law I am most likely to misuse is _____ because _____.
- A logarithm is an inverse because _____.
- A decimal threshold answer is not always the final answer because _____.
- One way I can independently verify a TI-84 result is _____.

End of Lesson 1.5 Full Notes

Exact structure first. Technology where useful. Verification always.